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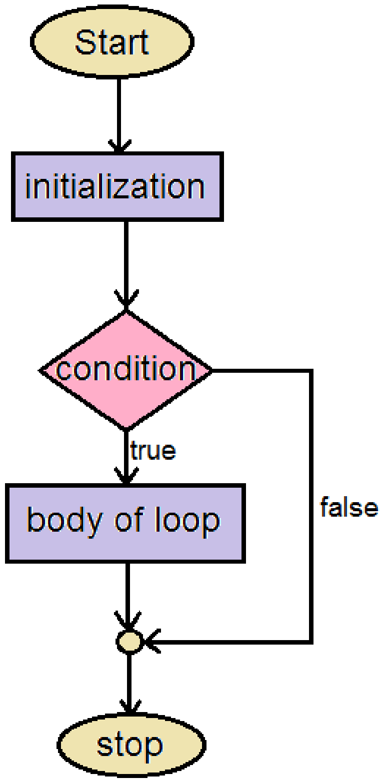
Iteration

Introduction to Computer Programming

Management and Artificial Intelligence

Iteration

- **Iteration** is the process where a set of instructions (or statements) is executed repeatedly for a specified number of time or until a condition is met.
- Like branching, these statements also **alter the control flow** of the program.



Anatomy of a looping mechanism.

while loops

```
while <condition>:
    statement
    statement
    ...
    statement
```

- <condition> evaluates to a Boolean (True or False)

    1. if <condition> is True, then execute all statements inside the while block
    2. evaluate <condition> again
    3. repeat until <condition> is False

Example

```python
n = input("You are in the Lost Forest. Go Left or Right?: ")
while n == "Right":
    n = input("You are in the Lost Forest. Go Left or Right?: ")
print("You are out of the Lost Forest.")
```

`for` loops

```
for <variable> in range(<number>):
    statement
    statement
    ...
    statement
```

- each time through the loop, `<variable>` takes some value
- `<variable>` starts from the smallest value (usually, `1`)
- next time, `<variable> = <variable> + 1`

The `range()` function

The `range(start, stop, step)` function returns a sequence of numbers, starting from `start`, and increments by `step`, and stops at `stop-1`. Keep in mind the following caveats:

- `stop` is mandatory
- `step` is optional: if not provided, it defaults to `1`
- `start` is optional: if not provided, it defaults to `0`

Examples:

```
mysum = 0
for i in range(7, 10):
    mysum = mysum + i
print(mysum)
```

```
24
```

```
mysum = 0
for i in range(5, 11, 2):
    mysum = mysum + i
print(mysum)
```

```
21
```

`for` vs. `while` loops

Iterate through sequence of numbers in `[0, 4)`:

- With a `while` loop:

```
n = 0
while n < 4:
    print(n)
    n = n + 1
```

```
0
1
2
3
```

- With a `for` loop:

```
for n in range(4):
    print(n)
```

```
0
1
2
3
```

Main differences

for loops

- **known** number of iterations
- can **end early** via `break`
- (natively) uses a **counter**
- you **can rewrite** a `for` loop using a `while` loop

while loops

- **unbounded** number of iterations
- can **end early** via `break`
- you can use a **counter**, but you must initialise it before the loop and increment it inside the loop
- you **may not be able to rewrite** a `while` loop using a `for` loop

More on `for` loops

`for` loops can be used to iterate over other data structures (e.g., lists or dictionaries*).

Example: print all functions in the `math` module that contain the word `'log'`;

```
import math
for f in dir(math):
    if 'log' in f:
        print(f)
print("Done!")
```

```
log
log10
log1p
log2
Done!
```

RECALL: `dir()` returns a list.

* We will see lists and dictionaries later in the course

Nested Loops

We know that branching (i.e., `if`) statements can be arbitrarily nested. The same holds for loops.

Example:

```
mysum = 0

for i in range(3):
    for x in range(2):
        mysum += i

print(mysum)
```

```
6
```

RECALL: `mysum += i` is the same as `mysum = mysum + i`

`break` statement

The `break` statement:

- immediately exits whatever loop is it in
- all further statements in the block are skipped
- exits only the innermost loop
- usually triggered at given condition

```
while <condition_1>:
    while <condition_2>:
        <statement_a>
        break           # exits inner loop
        <statement_b>   # never executed
    <statement_c>
```

Example

```
mysum = 0

for i in range(5, 11, 2):
    mysum += i
    if mysum == 5:
        break
    mysum += i

print(mysum)
```

```
5
```

What's going on here?

Nested loops with `break`, an example:

```
mysum = 0

for i in range(30):
    for n in range(20):
        if n > 2:
            break
        mysum += i
    if i>1:
        break

print(mysum)
```

```
9
```

What's going on here?

Warm-up exercises

- Notebook with exercises on iteration (no hints)
- Notebook with exercises on iteration (with pseudocode)
- Notebook with exercises on iteration (with solutions)

Exercise session

Find the exercises for the exam session: here